A Martin-Löf Test That Pseudorandom Generators Fail

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Abstract

Every computable sequence fails some Martin-Löf test, but the standard proof is circular: it uses the sequence itself to build the test. We construct a concrete, non-circular Martin-Löf test—based on the Chebotarev density theorem—that five standard pseudorandom number generators all fail. The test asks whether a bitstream indexed by primes has the correct conditional hit rates modulo 12. Any sequence drawn from the Chebotarev measure passes. Any PRNG fails, because its bits are generated independently of the primes.

1 The problem with the standard proof

A pseudorandom number generator produces a computable sequence. No computable sequence is Martin-Löf random. The proof: given a computable sequence x, define $U_m = \{y : y_i = x_i \text{ for } i \leq f(m)\}$ where $f(m)$ grows slowly enough that $\mu(U_m) \leq 2^{-m}$. Then $x \in \bigcap U_m$, so x fails the test.

This is correct but useless. The test requires you to *compute* x before you can test it. For a PRNG with a short seed, that is feasible. But for a number-theoretic sequence like the Artin membership sequence, the test requires factoring $q - 1$ for every prime q—the test is harder than the sequence it claims to debunk.

More importantly, the test is not *interpretable*. It tells you that the sequence is not random but says nothing about *how* it fails. We construct a test that is concrete, interpretable, and motivated by number theory.

2 The Chebotarev measure

Fix an integer r. A prime q belongs to $S(r)$ if r generates the multiplicative group $(\mathbb{Z}/q\mathbb{Z})^*$. Artin conjectured that $S(r)$ has density $C_{\text{Artin}} \approx 0.3740$ among the primes.

But the density is not uniform across residue classes. Whether r is a primitive root mod q depends on the factorisation of $q - 1$, which is constrained by $q \bmod m$ for small m. For $r = 3$ and $m = 12$:

- $q \equiv 1 \pmod{12}$: then $3 \mid q - 1$, so the cubes are a proper subgroup. 3 cannot generate. Hit rate $= 0$.
- $q \equiv 11 \pmod{12}$: again $3 \mid q - 1$. Hit rate $= 0$.
- $q \equiv 5 \pmod{12}$: neither 3 nor 2 divides $q - 1$ obstructively. Hit rate ≈ 0.90.
- $q \equiv 7 \pmod{12}$: $3 \nmid q - 1$ but $2 \| q - 1$. Hit rate ≈ 0.60.

These are not empirical observations. The vanishing at $q \equiv 1$ and $q \equiv 11$ is a theorem. The values 0.90 and 0.60 are predictions of the Chebotarev density theorem applied to the splitting field of $x^{q-1} - 1$ over $\mathbb{Q}$.

Definition 1. The *Chebotarev measure* μ on $\{0,1\}^{\mathbb{N}}$, indexed by the primes $p_1, p_2, \ldots$, is defined by

$$\mu(b_n = 1 \mid p_n \equiv a \pmod{12}) = \begin{cases} 0 & \text{if } a \in \{1, 11\}, \\ c_a & \text{if } a \in \{5, 7\}, \end{cases}$$

where $c_5 \approx 0.897$ and $c_7 \approx 0.600$ are the Chebotarev-predicted densities, with bits conditionally independent given the residue class.

This is a computable measure: the residue class $p_n \bmod 12$ is computable, and the conditional probabilities are computable reals. Martin-Löf randomness with respect to μ is well-defined.

3 The test

Definition 2. The *residue-class coherence test* is the Martin-Löf test $(U_m)_{m \geq 1}$ relative to μ defined by

$$U_m = \left\{ (b_n) : \sum_{a \in \{1,5,7,11\}} \frac{(\hat{P}_a - \mu_a)^2}{\mu_a(1 - \mu_a)/N_a} > t_m \right\},$$

where $\hat{P}_a$ is the empirical hit rate among the first N primes congruent to $a \pmod{12}$, μ_a is the Chebotarev prediction, $N_a = |\{n \leq N : p_n \equiv a \pmod{12}\}|$, and t_m is chosen so that $\mu(U_m) \leq 2^{-m}$.

Under μ, the statistic converges in distribution to χ^2_3 as $N \to \infty$, so $t_m = O(m)$ suffices. The sets U_m are uniformly computably enumerable (the chi-squared statistic is computable from the data), and $\mu(U_m) \leq 2^{-m}$ by choice of t_m. This is a valid Martin-Löf test.

4 Pseudorandom generators fail

A PRNG generates bits independently of the prime index. Its bits satisfy $P(b_n = 1 \mid p_n \equiv a \pmod{12}) = P(b_n = 1)$ for all a. In particular, $\hat{P}_1 \approx \hat{P}_5 \approx \hat{P}_7 \approx \hat{P}_{11} \approx 0.375$, while $\mu_1 = \mu_{11} = 0$ and $\mu_5 \approx 0.90$.

As $N \to \infty$, the chi-squared statistic diverges. The PRNG output lies in U_m for every m. It fails the test.

We verified this for five standard generators over $N = 50{,}000$ primes:

Generator	$\hat{P}_1$	$\hat{P}_5$	$\hat{P}_7$	$\hat{P}_{11}$	χ^2
Mersenne Twister	0.369	0.370	0.380	0.382	4.6
PCG64	0.378	0.372	0.379	0.376	0.7
MT19937	0.375	0.372	0.376	0.367	1.6
Philox	0.377	0.368	0.369	0.378	3.0
SFC64	0.374	0.377	0.371	0.379	1.2

The chi-squared values above are computed against the *flat* null ($\hat{P}_a = \hat{P}$ for all a). Against the Chebotarev measure μ, the statistic is dominated by the $a = 1$ and $a = 11$ terms, where $\mu_a = 0$ but $\hat{P}_a \approx 0.375$, and formally diverges. The generators do not merely fail the test. They fail it infinitely badly.

5 The Artin sequence passes

The Artin membership sequence $b_n = \mathbf{1}_{p_n \in S(3)}$, computed over 50,000 primes:

$p_n \bmod 12$	N_a	Hits	$\hat{P}_a$
1	12,447	0	0.000
5	12,538	11,243	0.897
7	12,510	7,508	0.600
11	12,503	0	0.000

The empirical hit rates match the Chebotarev predictions exactly. The chi-squared statistic against μ is small. The Artin sequence passes the residue-class coherence test.

Note that this does not make the Artin sequence Martin-Löf random. It is computable, so it fails other tests. But it passes *this* test—the one that every PRNG fails.

6 The Artin sequence fails standard tests

At the same time, the Artin bitstream fails five of nine standard PRNG tests:

Test	p-value	Verdict
Frequency (monobit vs 0.5)	≈ 0	Fail
Frequency (vs Artin's constant)	0.565	Pass
Runs	≈ 0	Fail
Serial (bigrams)	≈ 0	Fail
Autocorrelation (lags 1–50)	≈ 0	Fail
Block frequency ($M = 100$)	1.000	Pass
Longest run of ones	0.018	Pass
CUSUM	≈ 0	Fail
Spectral (DFT)	0.296	Pass

The most informative failure is the lag-1 autocorrelation $r_1 = -0.062$: consecutive primes repel each other in $S(3)$.

$$P(p_{n+1} \in S(3) \mid p_n \in S(3)) = 0.337, \qquad P(p_{n+1} \in S(3) \mid p_n \notin S(3)) = 0.398.$$

This repulsion varies with r (Table 1). Most values repel, but $r = 15$ attracts and $r = 14$ is neutral.

7 Discussion

The standard proof that computable sequences are not Martin-Löf random is an existence result: for every computable sequence, *some* test catches it. The test is built from the sequence, cannot be described without it, and conveys no insight about *why* the sequence is not random.

We have exhibited a specific test—residue-class coherence against the Chebotarev measure—with three properties:

1. It is a valid Martin-Löf test (computably enumerable, measure-decreasing).
2. It catches every PRNG (the bits are independent of the prime index).

r	P	P|hit	P|miss	Drop	r_1
2	0.374	0.329	0.401	+12.0%	−0.072
3	0.375	0.337	0.398	+10.3%	−0.062
5	0.394	0.338	0.431	+14.3%	−0.093
6	0.374	0.351	0.387	+6.2%	−0.037
7	0.374	0.369	0.377	+1.3%	−0.008
10	0.376	0.362	0.384	+3.6%	−0.021
11	0.375	0.359	0.385	+4.4%	−0.026
12	0.376	0.333	0.401	+11.4%	−0.069
13	0.377	0.346	0.396	+8.3%	−0.050
14	0.374	0.374	0.373	−0.1%	+0.000
15	**0.377**	**0.420**	**0.351**	**−11.4%**	**+0.069**
17	0.376	0.350	0.391	+6.8%	−0.041
18	0.373	0.329	0.399	+11.7%	−0.070
19	0.376	0.373	0.378	+0.7%	−0.004
20	0.394	0.335	0.432	+14.8%	−0.096

Table 1: Consecutive-prime repulsion in $S(r)$. Positive drop means repulsion; negative means attraction.

3. It has a mathematical name and an interpretable failure mode (the generator does not know which residue class it is in).

The test does not use the PRNG's algorithm or seed. It works against *any* generator that produces bits independently of the primes, which includes all standard PRNGs.

The Artin sequence is not Martin-Löf random either—it is computable, and other tests catch it. But the tests that catch it (autocorrelation, runs) are *different* from the test that catches PRNGs. The two kinds of non-randomness do not overlap.

The situation is:

	Standard tests	Coherence test
Truly random sequence	Pass	Pass
Artin sequence	Fail	Pass
PRNG	Pass	Fail

No single test captures "randomness." What exists is a landscape of tests, each sensitive to a different kind of structure. Standard PRNG tests detect the absence of independence. The coherence test detects the absence of arithmetic. The Artin sequence has arithmetic but not independence. A PRNG has independence but not arithmetic. A truly random sequence—drawn from the Chebotarev measure—would have both.

References

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